

Planet Wars RTS AI

IEEE CoG 2026 Results

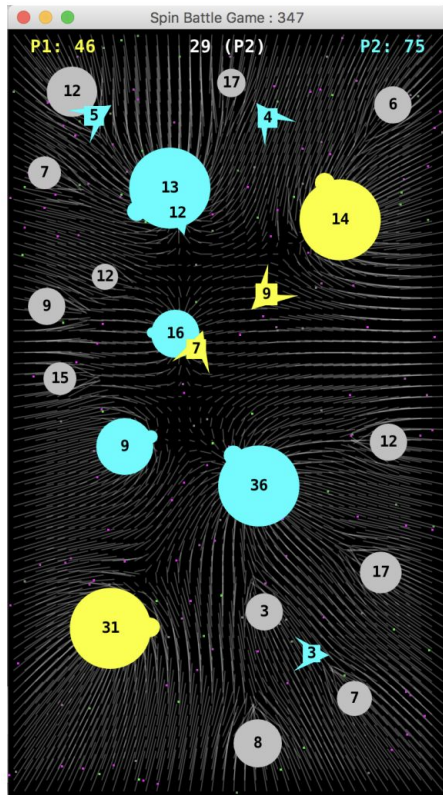
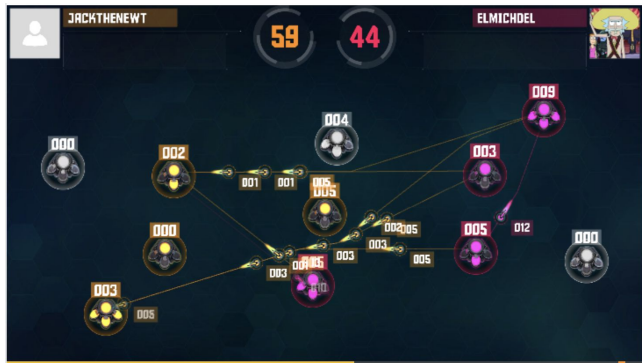
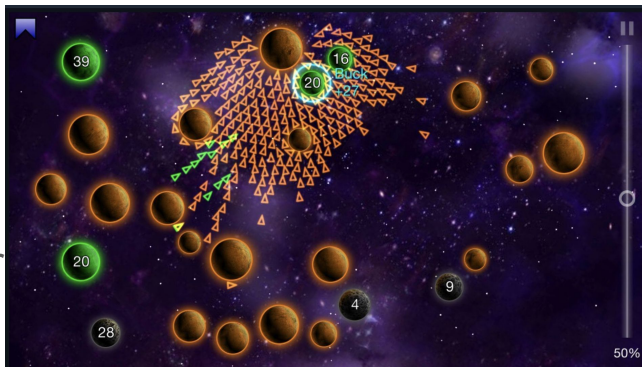
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History

- Galcon (2007-2008)
- Google AI Challenge Planet Wars (2010)
- Planet Wars at Dagstuhl Seminar (2015)
- Ghost in the Cell (2017) on CodingGame
- Fast Planet Wars Variants (Lucas, 2018 IEEE CIG)
 - Gravity, actuators, parameterised, fast forward models
- Planet-Wars RTS AI (Lucas, 2025)
- Orbit Wars, Kaggle 2026



Game Rules

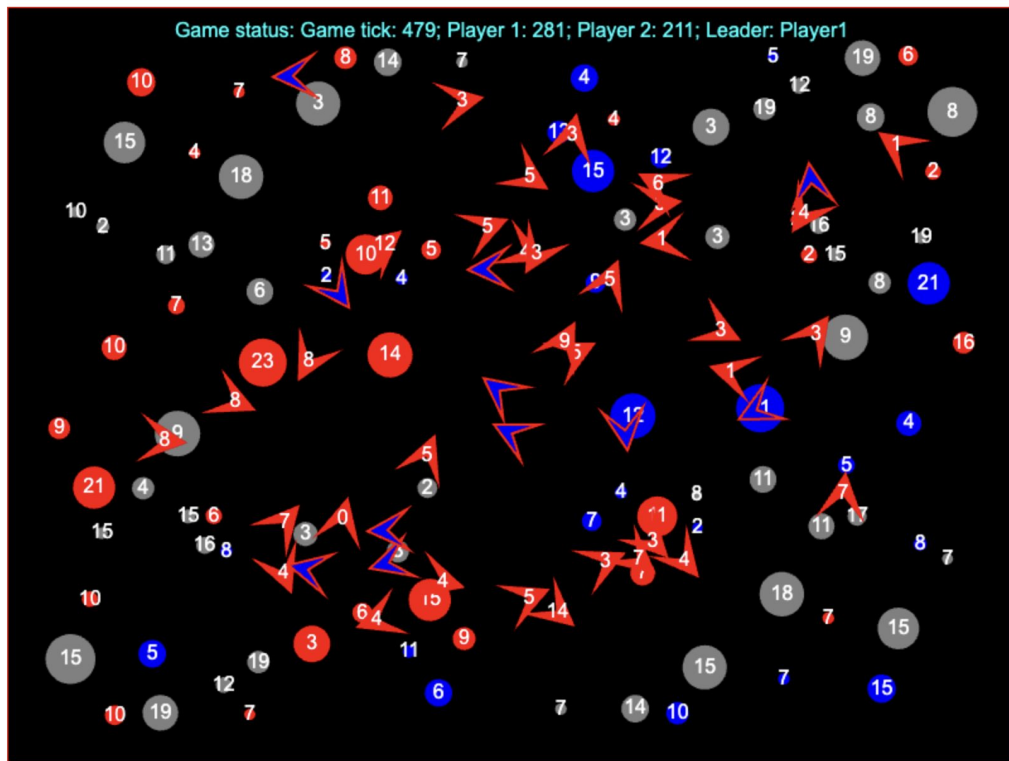
Win by eliminating opponent, or having more **ships on planets** when time runs out

Each planet has a single **transporter**

If you own the planet, and the transporter is at home, you can send ships across space - either to attack opponent or neutral planets, or to defend your own

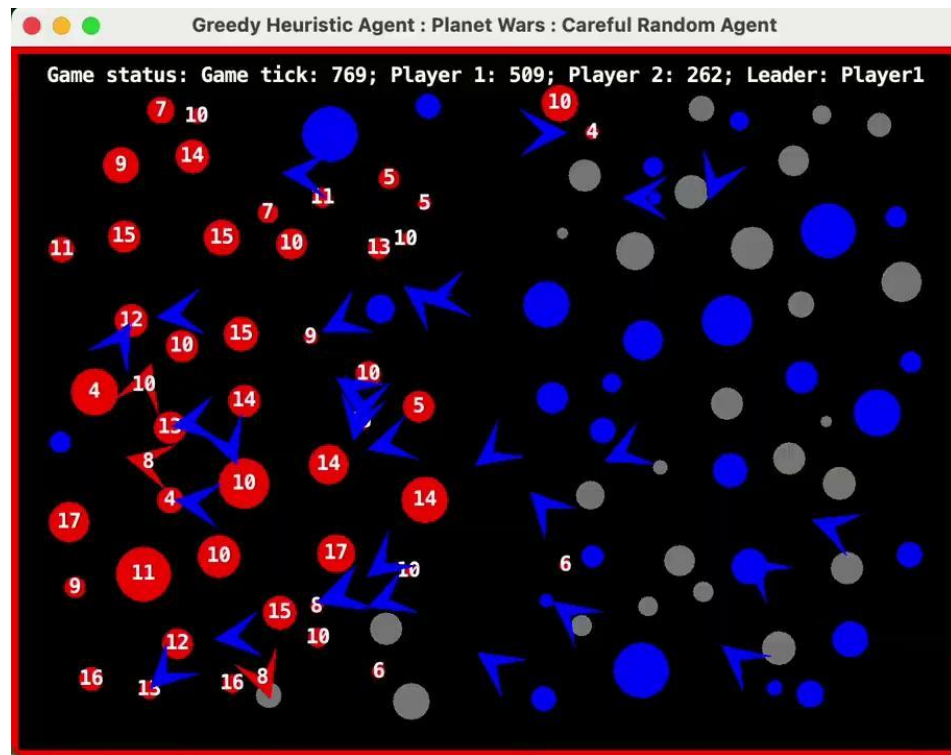
An invading force **a** will overcome the occupants **b** only if $a > b$ then ownership is transferred. Always $|a-b|$ ships remaining after battle.

Owned planets (not **neutral** ones) grow ships in proportion to their size



Another Game AI competition?

- Sweet spot for Game AI Challenges:
- RTS - **with real-time enforced**
- Highly configurable
- Hard test for sophisticated AI agents
- Low compute budget
- Challenge for RL agents to cope with high variability in maps and game params
- Open-ended challenge: **depends on opponents**
- **Fast Plannable Forward Model**
 - 1 million steps per second



[Video Link](#)

Some Interesting Research Questions

- Can Claude Code write a good agent?
- Can Evolutionary Program Search Design good agents?
 - LLaMEA
 - LLM4AD
 - EASE
 - DeepEvolve
 - OpenEvolve
 - ShinkaEvaolve
 - ... **DIY?**
- Deep RL?
- Graph Neural Networks?
- MCTS, Rolling Horizon Evolution, SFP Variants? (online planning)
- Human-design???

CoG 2026 Variations

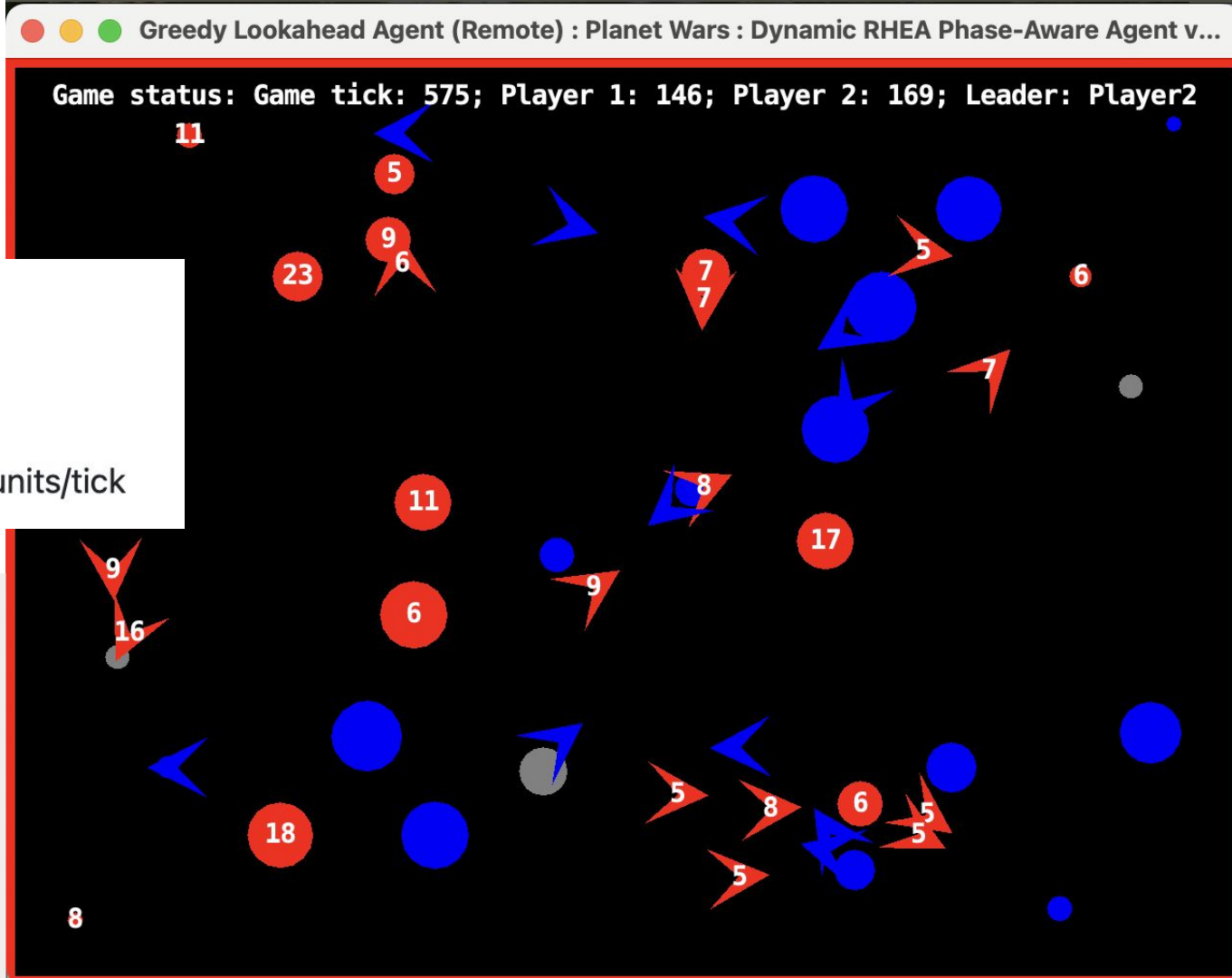
- Number of Planets: 10-30
- Neutral Ratio: 0.25-0.35
- Growth Rate: 0.05-0.2
- Transporter Speed: 2.0-5.0 units/tick

Branching factor:

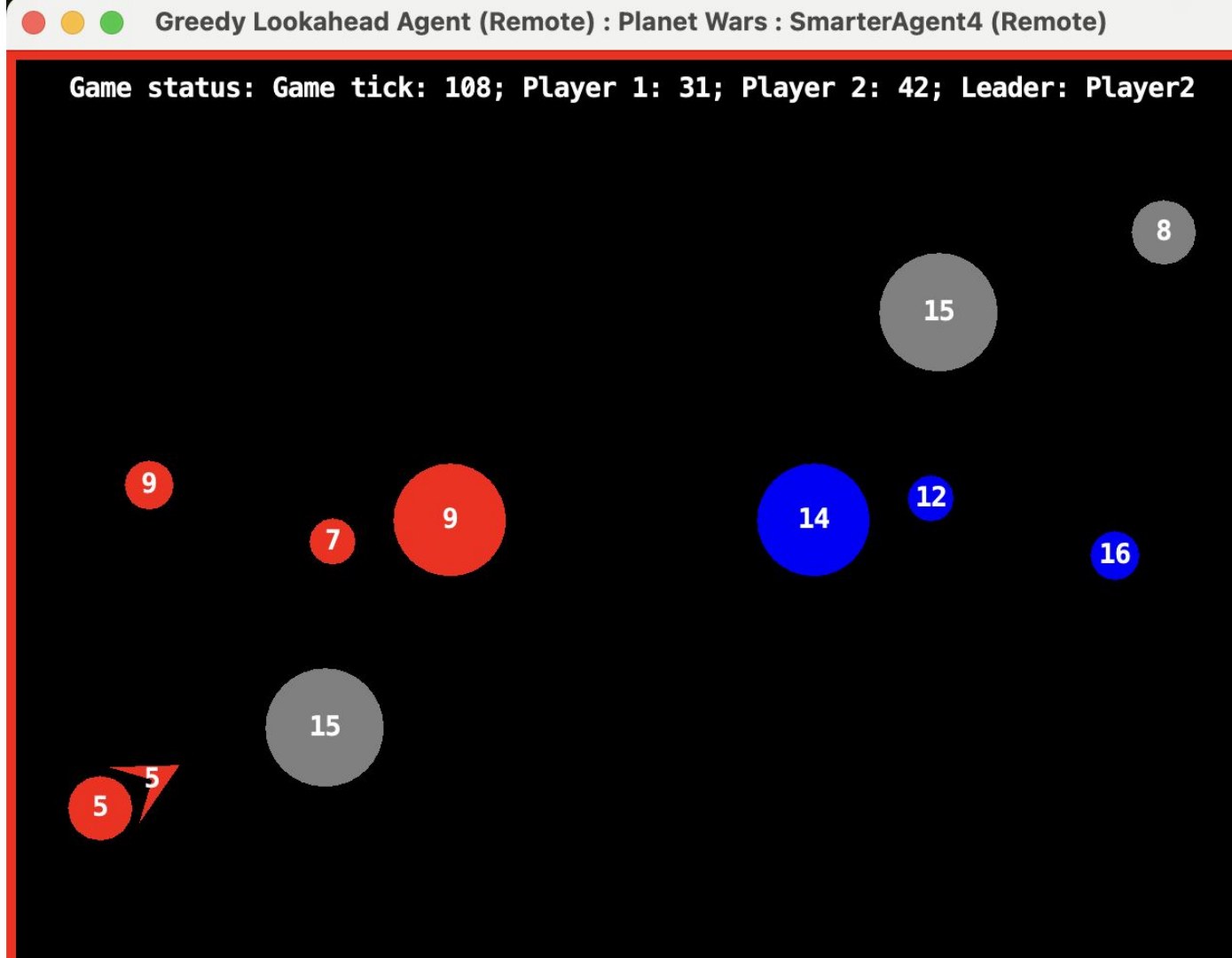
$$\text{nSource} * \text{nTarget} * \text{nShipOptions}$$

E.g. $10 \times 20 \times 10 = 2000$

Often much smaller



Small
number of
planets, fast
ships



AAMAS 2026 Ranking (top 12) - TrueSkill(™) League

Planet Wars Spring 2026 — Leaderboard

#	Agent	μ	σ	$\mu - 3\sigma$	Matches	Updated
1	teamtitanagentv3-256baf2	34.062	0.605	32.248	16040	2026-04-06 08:48
2	gnn_cont999_128_top64-97050fd	32.351	0.517	30.799	13900	2026-04-06 08:48
3	nanaboshi08271-5ea4531	29.686	0.572	27.968	11480	2026-04-06 08:48
4	nanaboshi08281-8df146e	29.145	0.528	27.559	11780	2026-04-06 08:48
5	nanaboshi08280-c2aa68b	28.693	0.515	27.149	11380	2026-04-06 08:48
6	nanaboshi08268-54e82c1	28.629	0.521	27.067	11140	2026-04-06 08:48
7	nanaboshi08270-b40ea88	28.473	0.506	26.954	11340	2026-04-06 08:48
8	gnn_cont999_200_top4-5027e43	27.598	0.502	26.091	11674	2026-04-06 08:48
9	gnn_cont999_200-a31c911	27.038	0.510	25.509	11734	2026-04-06 08:48
10	gnn_cont999_128_base-9531e0d	26.843	0.515	25.299	10160	2026-04-06 08:48
11	adversarial-dba-ba51c9e	23.311	0.501	21.807	8320	2026-04-06 08:48
12	galacticarmada-ecd6ae8	22.116	0.522	20.550	8720	2026-04-06 08:48

WCCI
2026
League
(June)

#	Agent	μ	σ	$\mu - 3\sigma$	Matches	Updated
1	anti-1-8c8bf1c	38.210	0.684	36.159	5140	2026-06-23 09:09
2	anti-0-b1b8324	34.435	0.631	32.541	4420	2026-06-23 09:09
3	teamtitanagentv3-256baf2	33.957	0.543	32.330	17460	2026-06-23 09:09
4	gnn_cont999_128_top64-97050fd	32.378	0.543	30.750	16600	2026-06-23 09:09
5	nanaboshi08281-8df146e	29.807	0.530	28.216	13380	2026-06-23 09:09
6	nanaboshi08280-c2aa68b	28.658	0.535	27.054	13140	2026-06-23 09:09
7	nanaboshi08271-5ea4531	28.417	0.540	26.796	13040	2026-06-23 09:09
8	nanaboshi08270-b40ea88	27.785	0.529	26.198	12760	2026-06-23 09:09
9	nanaboshi08268-54e82c1	27.561	0.537	25.950	12600	2026-06-23 09:09
10	gnn_cont999_200-a31c911	27.446	0.523	25.876	13094	2026-06-23 09:09
11	gnn_cont999_200_top4-5027e43	27.374	0.531	25.782	12994	2026-06-23 09:09
12	gnn_cont999_128_base-9531e0d	26.355	0.520	24.796	11460	2026-06-23 09:09
13	adversarial-dba-ba51c9e	25.965	0.519	24.408	9900	2026-06-23 09:09
14	planetwarriors-agent-v2-1-2efb60d	24.968	0.527	23.387	4740	2026-06-23 09:09
15	genetic_v1-04b0974	24.145	0.484	22.693	740	2026-06-23 09:09
16	galacticarmada-ecd6ae8	24.114	0.504	22.603	10100	2026-06-23 09:09

CoG

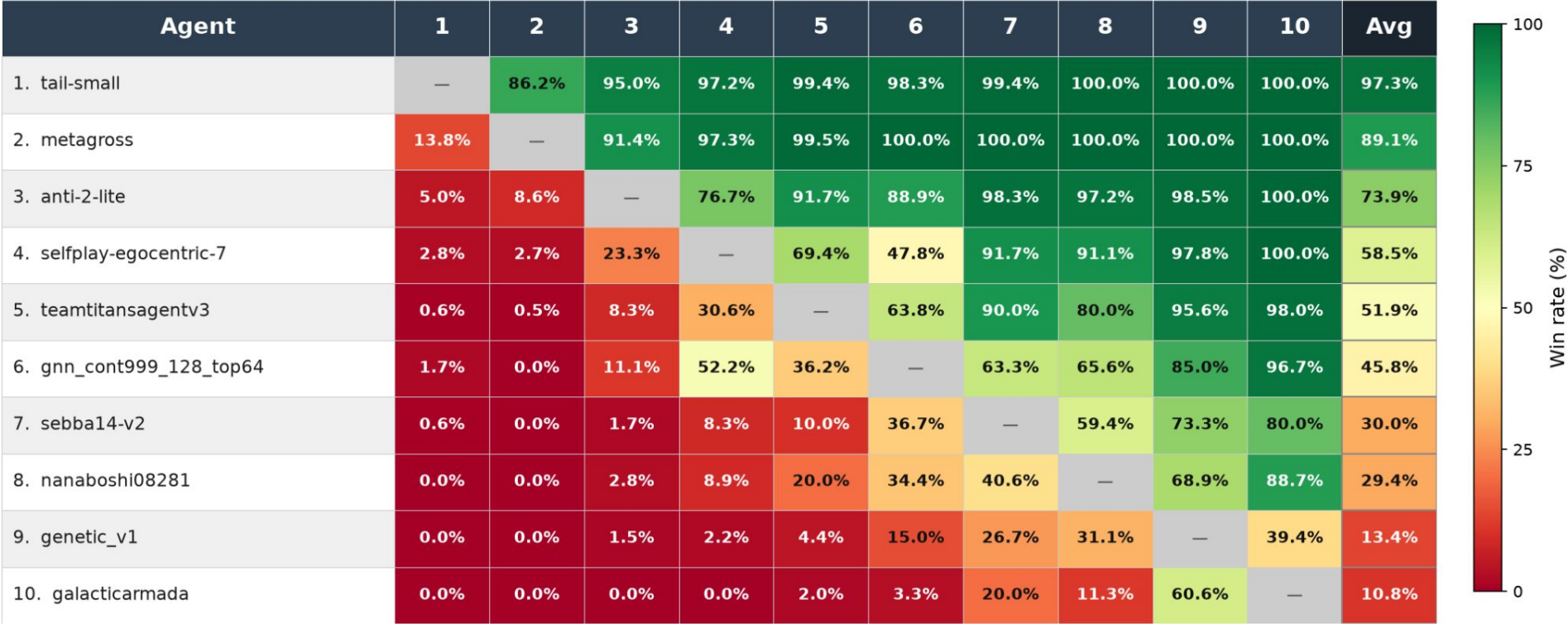
#	Agent	μ	σ	$\mu - 3\sigma$	Matches	Updated
1	tail-small-2e3c0f7	47.900	0.532	46.305	460	2026-08-31 17:30
2	metagross-efaebe5	46.857	0.567	45.156	460	2026-08-31 17:30
3	metang-6ac59b9	46.547	0.544	44.915	1000	2026-08-31 17:30
4	tail-warm-start-970524b	45.093	0.526	43.515	440	2026-08-31 17:30
5	beldum-2b51d65	43.088	0.515	41.543	1000	2026-08-31 17:30
6	tail-test-90e2323	42.820	0.512	41.285	440	2026-08-31 17:30
7	onix-39d1f88	39.422	0.483	37.973	880	2026-08-31 17:30
8	anti-2-lite-aabb557	39.373	0.511	37.839	420	2026-08-31 17:30
9	anti-2-a0db115	39.244	0.491	37.772	900	2026-08-31 17:30
10	wolfes-incineroar-2fd1340	39.102	0.491	37.630	2540	2026-08-31 17:30
11	mega-heracross-dbf7d66	39.038	0.487	37.578	900	2026-08-31 17:30
12	masked-royal-4ba9aab	38.406	0.495	36.921	2440	2026-08-31 17:30
13	incineroar-z-863250b	37.265	0.498	35.772	2320	2026-08-31 17:30
14	anti-1-8c8bf1c	35.433	0.498	33.939	6440	2026-08-31 17:30

Top 10 entries from distinct teams

Summary Table

Rank	Agent	GitHub repo	Language	Trained weights?	Planning / search?
1	tail-small-2e3c0f7	mziele1/planet-wars-submission	Python (JAX)	Yes — PPO checkpoint	No — pure NN forward pass
2	metagross-efaebe5	mnn31/planet-wars-agent	Kotlin	No	Yes — RHEA
3	anti-2-lite-aabb557	daniel-geerdink/planet-wars-rtg-research	Kotlin	No	Yes — forward sim + heuristic eval
4	selfplay-egocentric-7-2b99083	josephsarafidis/planet-wars-rtg-submission	Python (PyTorch)	Yes — GNN PPO	No — pure NN forward pass
5	teamtitanagentv3-256baf2	abuzar-rasool/planet-wars-rtg	Kotlin	No	Yes — forward sim
6	gnn_cont999_128_top64-97050fd	Priwinn/planet-wars-private	Python (PyTorch)	Yes — trained GNN	Yes — one-step look-ahead + value net
7	nanaboshi08281-8df146e	YuriCat/PlanetWarsSubmission	C++ / PyBind11	Yes — policy + value (param.bin)	Yes — multi-threaded MCTS
8	sebba14-v2-9d3b196	Sebba14/planet-wars-az-agent	Kotlin	No	No — greedy heuristic
9	genetic_v1-04b0974	likang03/planet-wars-rtg	Kotlin	No	Yes — evolutionary (SimpleEvoAgent)
10	galacticarmada-ecd6ae8	nikivanstein/planet-wars-rtg-python	Python	No	No — multi-heuristic reactive

CoG 2026 — Round-Robin Win Rates (row vs column)



2026 Results

Significant progress in run-up to IEEE CoG in Sep 2026

- AAMAS (May) - Team Titans: winner from 2025
- WCCI (June) - Anti - Heuristics and planning, LLM-evolved code
- EgoCentric (CoG) Graph NN, PPO
- Metagross - Rolling Horizon Evolution plus smart heuristics
- Tail-small - Vibe coded RL architecture trained through self play (around a day of training for each agent), with significant human expertise in the vibing!

Many different methods can be competitive, the devil is in the detail!

See linked slides for more info

Summary

- Congrats to the CoG 2026 winner! - this is not an easy challenge!
- Set it for your student projects!
- **Games are Living Benchmarks**
- **Next Edition: TBA, league will continue to run**
- **Competition runner code all open source**

